



TNX CODATHON 2K26



Team Name : Fin Fusion

Members Name & Roles:

- Avika Srivastava : UI/UX designer
- Ayushi Yadav : Analytics & Deployment / AI & Voice integration
- Gauri Mishra : Backend
- Gayatri Bose : Backend / AI & Voice integration
- Harshika Dixit (Leader) : Frontend / AI & Voice integration

Track : Fintech / Artificial Intelligence

VoiceKhata

An AI-Powered Voice-First Digital Ledger for Small Merchants



Problem Statement

The Problem :

Millions of small merchants across India, including kirana stores, tea stalls, pharmacies, and street vendors, still rely on handwritten notebooks (Khata) to record customer credit transactions.

Although digital ledger applications exist, many require typing, reading, and navigating multiple screens, making them difficult for merchants with limited digital literacy.

Challenges Faced :

- Manual bookkeeping is slow and prone to errors.
- Paper records can be lost or damaged.
- Typing transactions during busy business hours is inconvenient.
- Existing apps provide limited support for local languages and natural speech.
- Tracking customer balances becomes difficult over time.
- Anyone can manipulate the ledger entries, which may result in financial losses for the vendor and their business

Our Goal/ Aim of our project :

To build a simple voice-first digital ledger that allows merchants to record transactions naturally by speaking in their preferred language, without typing or complicated interfaces.



Solution Overview

VoiceKhata

VoiceKhata is an AI-powered voice-enabled bookkeeping assistant that converts spoken transactions into organized digital ledger entries.

Instead of typing, the merchant simply says:
"Ramesh ne ₹200 ka samaan udhaar liya "

The application automatically:

- Converts speech into text
- Identifies the customer's name
- Extracts the transaction amount
- Detects whether it is credit or payment
- Updates the customer's ledger instantly
- Recalculates the remaining balance

Key Features

- Voice-based transaction entry
- Automatic customer detection
- Smart ledger management
- Running balance calculation
- Transaction confirmation before saving
- Voice/Text balance enquiry

And all this can be easily done in Hindi-English mixed speech.

Innovation

Unlike traditional bookkeeping apps that rely on typing, VoiceKhata is designed around **natural human conversation**, making digital bookkeeping accessible even for merchants with limited digital skills.

Technology Stack

Frontend



 HTML5
 Tailwind CSS
 JavaScript .

Backend



 Python
 FastAPI

Handles API requests,
transaction processing, and
ledger management.

Speech Recognition



◆ Web Speech API / Whisper

Converts spoken Hindi-
English conversations into
text in real time.

Text Processing



 Rule-Based NLP
 Python Regex

Extracts: Customer
Name, Amount ,
Transaction Type,
Additional Notes, etc
from natural speech.

UI/UX Design



 Figma

Interactive prototype and
user interface design

Development Tools



 Visual Studio Code
 GitHub
 Figma



Implementation Roadmap

1. Problem Analysis

Identify challenges faced by small merchants.

2. Design

Create UI and user flow in Figma.

3. Development

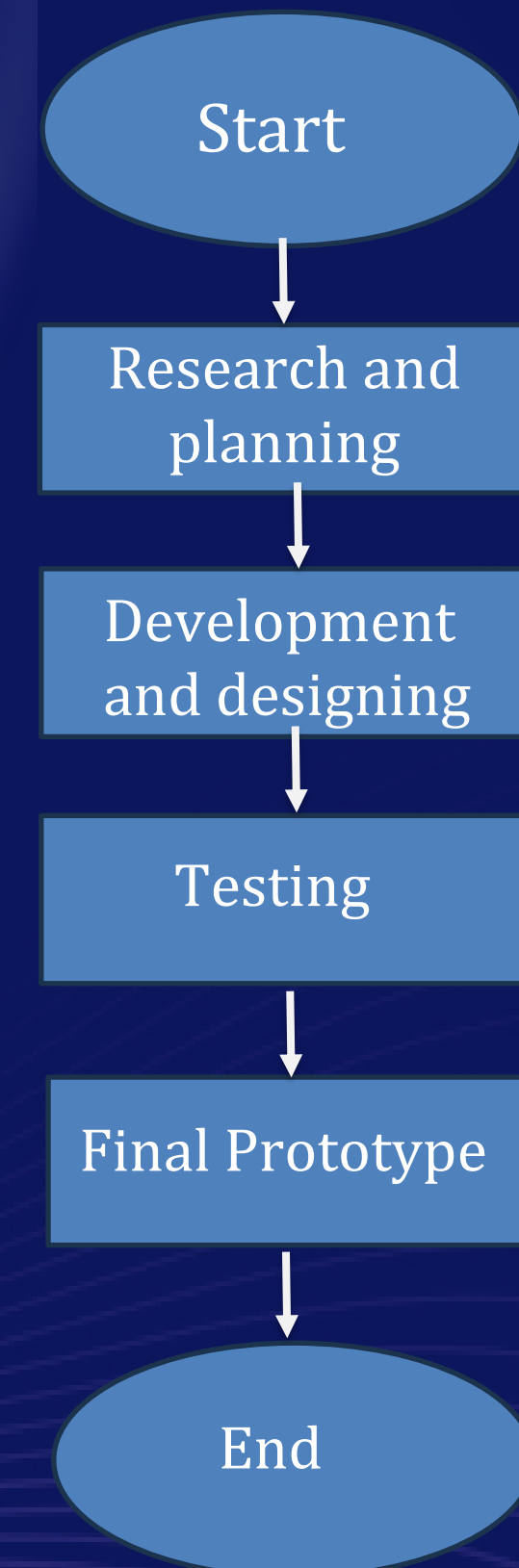
Build the frontend and backend.
Integrate voice recognition.

4. Testing

Test voice input and ledger updates.
Fix bugs and improve accuracy.

5. Final Prototype

Prepare the working demo and presentation.



Expected Impact



Target Users

Kirana Stores
Grocery Shops
Tea Stalls
Pharmacies
Small Retail Businesses
Street Vendors



Business Impact

Faster transaction recording
Reduced manual bookkeeping errors
Easier customer balance management
Improved operational efficiency



Social Impact

- Promotes financial inclusion
- Supports merchants with low digital literacy
- Reduces dependence on paper records
- Encourages digital adoption among small businesses



Expected Outcome

VoiceKhata simplifies bookkeeping by allowing merchants to continue working the way they naturally communicate, making digital finance more accessible and practical.



Future Scope

Although our hackathon focuses on the Minimum Viable Product (MVP), VoiceKhata has strong potential for future expansion.

Planned Enhancements

- Voice-based balance enquiry
- WhatsApp chatbot integration
- Multi-language support
- Smart payment reminders
- Daily transaction summaries
- Analytics dashboard
- Inventory management
- OCR-based bill scanning
- Offline functionality
- UPI payment integration

Long-Term Vision

To become a complete AI-powered financial assistant for India's small merchants, helping them manage bookkeeping, payments, and business insights through simple voice interactions.



Prototype Preview

Voice AI

Ask anything about your business

Ask AI

+ Record

RK

VoiceKhata AI

• Online • Hindi • English

Namaste! 🙏 Main VoiceKhata AI hoon.

Aap mujhse kisi bhi customer ka balance, sales summary, ya koi bhi business information pooch sakte hain. Hindi ya English — dono mein jawab dunga.

Hello! Ask me anything about your business in Hindi or English.

10:30 AM

Ramesh ka kitna udhaar hai?

10:31 AM

Ramesh Kumar — Outstanding Balance

🔥 Current Dues: ₹2,850

📅 Last transaction: 4 Aug 2026

🛒 Last purchase: Cheeni, Atta (₹500 credit)

📄 Total credit given: ₹18,400

✅ Total paid: ₹15,550

💡 Suggestion: 4 din se payment nahi aaya — reminder bhejne ka sahi waqt hai.

10:31 AM

Aaj ki total sales kitni hai?

10:32 AM

Aaj ki Sales — 4 Aug 2026

🟢 Total Sales: ₹12,450

📄 Transactions: 23

💰 Cash Collected: ₹7,800

🔥 Credit Given: ₹4,650

Kal se 18% zyada hai — bahut badhiya! 🎉

Is hafte ka best day abhi tak!

10:32 AM

Quick Queries

Sunita ka balance?

Is hafte ki report

Top 5 customers

Pending payments

Aaj ka profit?

New customer add karo

You can ask:

✓ Customer balance in Hindi

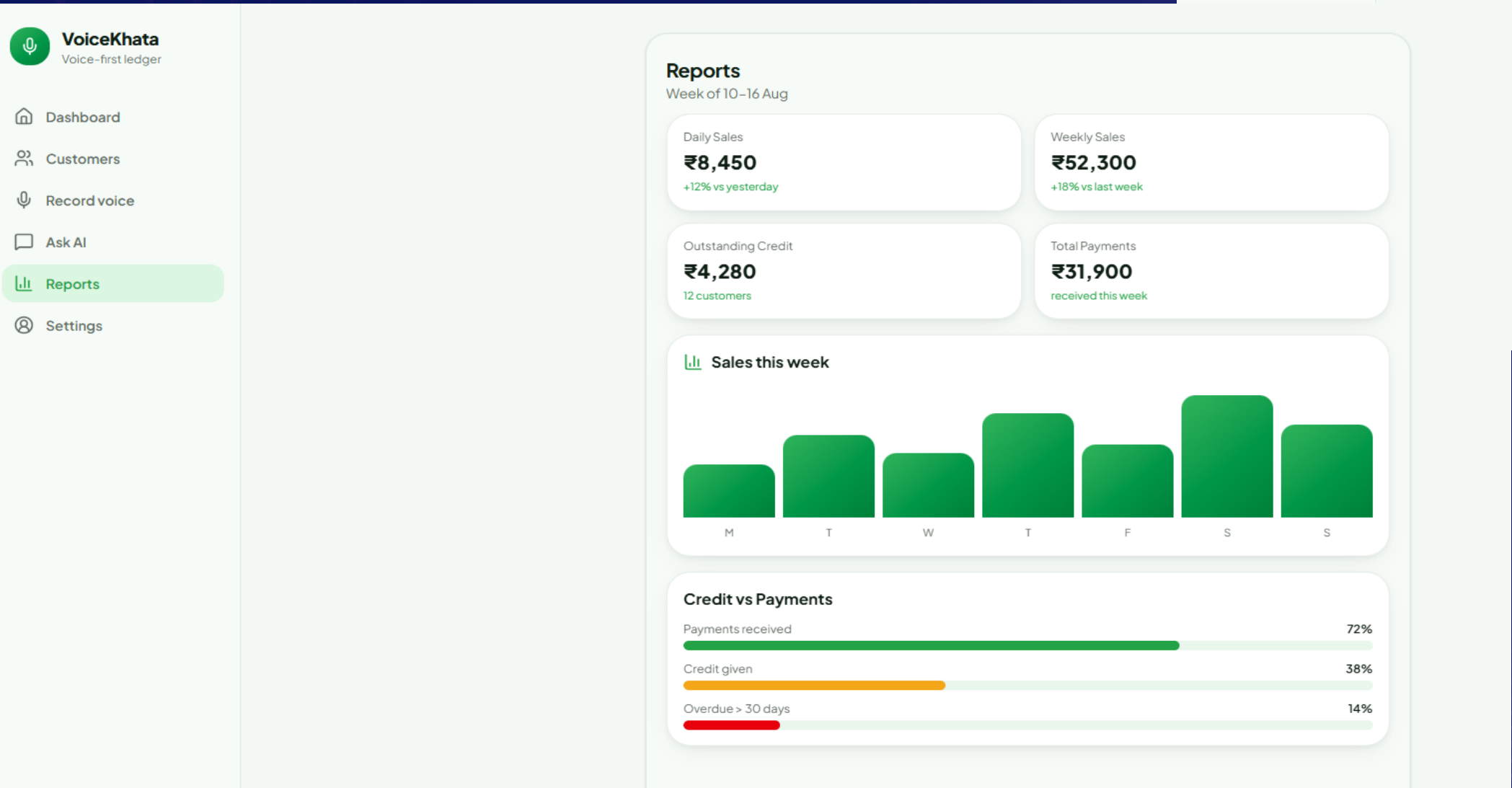
✓ Daily / weekly totals

✓ Overdue reminders list

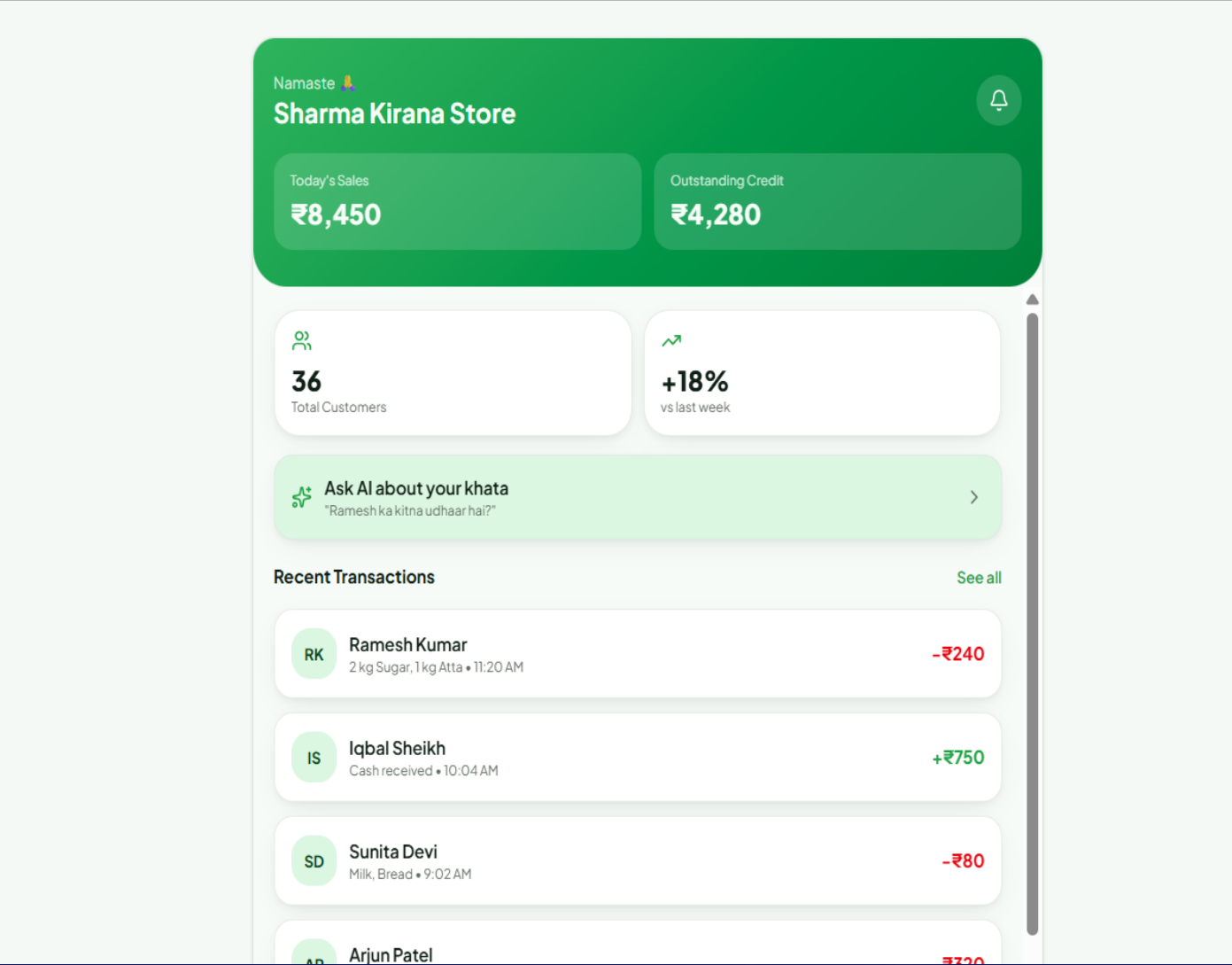
✓ Best-selling items

Kuch poochein... / Ask anything in Hindi or English

AI Business Assistant



Reports and analytics



Dashboard